SPECIFICATION — MERIDIA DUAL-PROTOCOL & SPINE ARCHITECTURE

SESSION-BASED DUAL-PROTOCOL SENSING (PASSIVE/ACTIVE) AND CONFORMING SPINE MECHANICAL ARCHITECTURE FOR INTRALUMINAL MAPPING (NON-LIMITING)

A. Technical Field

This disclosure relates to intraluminal physiological mapping devices and, more particularly, to: (1) a dual-protocol sensing methodology comprising a passive baseline mapping protocol (v1) and an active micro-vibration response protocol (v2) for characterizing internal tissue geometry, compliance, and reactive behavior; and (2) a conforming spine mechanical architecture for intraluminal insertable devices, including spine design variants, placement rules, and zone-to-spine decoupling requirements. This disclosure supplements the Node 4 disclosures (Documents 52-54) with specific protocol sequencing and mechanical backbone architecture.

B. Background and Motivation (Non-Limiting)

The existing Node 4 disclosures describe an intraluminal device with controlled radial stimulus (inflation/expansion) for measuring internal geometry and compliance. However, they do not disclose: (a) a structured dual-protocol approach in which a purely passive mapping session precedes and establishes baseline for an active micro-vibration-based session; (b) differential analysis methods comparing passive and active maps to reveal reactive tissue regions; or (c) specific conforming spine architectures that govern sensor zone placement, mechanical decoupling, and measurement accuracy. The present disclosure addresses these gaps.

C. Dual-Protocol Overview (Non-Limiting)

In certain embodiments, the intraluminal mapping device operates in two distinct sensing protocols performed in a defined sequence:

1. Protocol v1 — Passive Baseline Mapping: A session conducted with zero active stimulus (no vibration, no inflation). The device passively senses contact pressure distribution, internal geometry via distance sensors, temperature gradients, and impedance characteristics. The output is a baseline map representing the resting-state geometry and pressure distribution of the internal environment.
2. Protocol v2 — Active Micro-Vibration Response Mapping: A session conducted after v1 baseline acquisition in which controlled micro-vibration stimulus is applied while simultaneously measuring contact pressure, displacement, and impedance response. The output is an activation-response map showing how tissue responds to controlled mechanical input.

Protocol v1 must complete successfully and produce a validated baseline map before Protocol v2 may be initiated. This sequencing ensures that active-response data is always interpretable against a stable passive baseline.

D. Protocol v1 — Passive Baseline Mapping (Non-Limiting)

In various embodiments, Protocol v1 acquires the following measurements in a purely passive mode:

- Pressure Distribution Map: Multi-zone circumferential pressure sensing captures the resting contact pressure pattern across all instrumented zones along the device length
- Distance/Geometry Profile: Time-of-flight or IR distance sensors measure internal diameter or cavity geometry at multiple positions
- Temperature Gradient: Distributed temperature sensors capture the thermal profile along the device, providing context and confirming proper placement
- Impedance Baseline: Bio-impedance measurements establish baseline tissue electrical characteristics at each electrode pair
- IMU Stability: Inertial measurements confirm device stability and flag motion artifacts

The v1 output comprises a validated baseline pressure map, a geometry profile, quality indices (contact integrity, stability, temperature plausibility), and a confidence indicator. Segments failing quality criteria are rejected, and the user may be prompted to re-seat the device.

E. Protocol v2 — Active Micro-Vibration Response (Non-Limiting)

In various embodiments, Protocol v2 applies controlled micro-vibration stimulus while continuing to measure all sensors. The micro-vibration subsystem comprises:

1. Stimulus Source: One or more miniature LRA or piezoelectric actuators embedded within the device body, configured to produce controlled, low-amplitude mechanical vibration at specified frequencies.
2. Stimulus Isolation: Mechanical isolation structures (elastomeric decouplers, damping layers) prevent vibration from propagating directly into pressure and distance sensors, ensuring that measured responses reflect tissue reaction rather than direct mechanical coupling.
3. Frequency Sweep: The stimulus may be applied at a single frequency or swept across a frequency range to characterize frequency-dependent tissue response (e.g., stiffness, damping, resonance).
4. Zone-Sequential Activation: In embodiments with multiple actuators, stimulus may be applied to one zone at a time while other zones measure passive response, enabling spatial mapping of reactive regions.
5. Concurrent Sensing: Throughout v2 stimulus, all passive sensors continue to operate, capturing how pressure distribution, distance measurements, and impedance change in response to micro-vibration.

F. Differential Analysis — Passive vs. Active (Non-Limiting)

In certain embodiments, the system performs differential analysis by comparing v1 baseline maps with v2 activation-response maps. This analysis reveals:

- Reactive Regions: Zones where pressure distribution changes significantly between v1 and v2, indicating tissue areas with high mechanical responsiveness
- Compliance Mapping: Zones where distance/geometry measurements shift under micro-vibration, indicating relative tissue stiffness or elasticity
- Impedance Response: Zones where bio-impedance changes under mechanical stimulus, potentially indicating differences in tissue hydration, density, or composition
- Damping Characteristics: The rate at which tissue response decays after stimulus cessation, providing information about viscoelastic properties
- Frequency-Dependent Behavior: How tissue response varies with stimulus frequency, enabling characterization of mechanical resonance properties

Differential outputs are expressed as relative change indices, zone-based response maps, and confidence indicators. No absolute anatomical measurements are exposed.

G. Conforming Spine Architecture — Overview (Non-Limiting)

In various embodiments, the intraluminal device includes a mechanical backbone (spine) that provides structural support, sensor positioning, and controlled flexibility. The spine architecture must satisfy competing requirements: rigidity sufficient to maintain sensor positioning, flexibility sufficient to conform to internal geometry, and mechanical decoupling sufficient to prevent the spine from corrupting pressure measurements.

H. Spine Design Variants (Non-Limiting)

In various embodiments, the spine may be implemented using one of the following architectures:

1. Continuous Flex Backbone: A single continuous flexible element (e.g., a spring-steel or nitinol strip, a flexible PCB, or a silicone-encapsulated flex cable) running the length of the device. Advantages include smooth bending, predictable flex characteristics, and manufacturing simplicity. Disadvantages include potential for spine-induced pressure artifacts and limited ability to decouple individual zones.
2. Segmented Spine: A series of rigid or semi-rigid segments connected by controlled articulation points (e.g., living hinges, ball-and-socket joints, or elastomeric links). Advantages include zone-level mechanical independence, better decoupling of pressure measurements from spine behavior, and the ability to conform to complex geometries. Disadvantages include greater manufacturing complexity and potential for inter-segment wear.
3. Hybrid Spine: A combination of continuous flex regions and segmented regions along the device length, optimizing flexibility and decoupling for different functional zones.

I. Spine Placement Rules and Zone-to-Spine Decoupling (Non-Limiting)

In various embodiments, the relationship between the spine and sensor zones is governed by strict placement rules:

1. Radial Offset: Pressure-sensitive zones must be radially offset from the spine axis to minimize spine-induced contact pressure artifacts. The minimum offset distance is determined by the spine stiffness and the pressure sensor sensitivity threshold.
2. Mechanical Decoupling: Each pressure-sensing zone includes an isolation structure (e.g., elastomeric cushion, floating mount, or compliant membrane) between the zone and the spine, preventing spine bending forces from registering as contact pressure.
3. Angular Distribution: Sensor zones are distributed circumferentially such that no zone is directly loaded by spine bending in the primary flex direction.
4. Zone Independence: In segmented spine designs, each sensor zone is associated with an individual spine segment, and inter-segment articulation boundaries are positioned between zone boundaries to prevent cross-zone mechanical coupling.
5. Vibration Path Isolation: For Protocol v2, the micro-vibration stimulus path from actuator to tissue must bypass the spine or be decoupled from it to prevent spine resonance from generating measurement artifacts.

J. Spine-Pressure Interaction Calibration (Non-Limiting)

In certain embodiments, the system includes a spine-pressure interaction calibration procedure:

- Factory Calibration: The device is calibrated during manufacturing by applying known external pressures at each zone position and measuring the spine-induced baseline offset
- In-Session Calibration: During Protocol v1, the system may identify and subtract spine-induced pressure components by analyzing the correlation between device curvature (measured by IMU) and zone-level pressure readings
- Curvature Compensation: A compensation model maps device bend radius (estimated from IMU and/or spine strain sensing) to expected spine-induced pressure artifacts at each zone, subtracting these from the measured contact pressure

K. Exemplary Claim-Like Statements (Non-Limiting)

A provisional application does not require claims. The following statements are illustrative only and do not limit the disclosure.

1. A system comprising: a) an insertable intraluminal device having a plurality of pressure sensing zones, one or more distance sensors, and one or more micro-vibration actuators; b) a controller configured to execute a first protocol (v1) in which the device acquires a passive baseline map with zero active stimulus, and a second protocol (v2) in which the device applies controlled micro-vibration stimulus while continuing to acquire sensor data; and c) a processing module configured to perform differential analysis between v1 and v2 outputs to identify reactive regions, compliance characteristics, and frequency-dependent tissue response.
2. The system of statement 1, wherein the controller enforces sequential protocol execution such that v2 cannot be initiated until v1 has produced a validated baseline map meeting defined quality criteria.
3. The system of statement 1, wherein micro-vibration stimulus is mechanically isolated from pressure sensing zones by decoupling structures preventing direct mechanical coupling between actuators and sensors.
4. The system of statement 1, wherein the micro-vibration stimulus comprises a frequency sweep for characterizing frequency-dependent tissue response including stiffness, damping, and resonance.
5. The system of statement 1, further comprising a conforming spine providing structural support and sensor positioning, wherein the spine is mechanically decoupled from pressure sensing zones by isolation structures preventing spine bending forces from registering as contact pressure.
6. The system of statement 5, wherein the spine comprises one of: a continuous flex backbone, a segmented spine with controlled articulation points, or a hybrid combination thereof.
7. The system of statement 5, further comprising a curvature compensation module configured to estimate device bend radius from inertial measurements and subtract spine-induced pressure artifacts from contact pressure readings at each zone.
8. A method comprising: a) inserting an intraluminal device and executing a passive baseline mapping protocol

with zero active stimulus to produce a validated baseline pressure map and geometry profile; b) executing an active micro-vibration protocol applying controlled stimulus while continuing to acquire sensor data; c) performing differential analysis between the passive baseline and active response data to identify reactive tissue regions and compliance characteristics; and d) outputting privacy-preserving zone-based response maps and relative change indices.

9. The method of statement 8, wherein differential analysis includes computing reactive region indices, damping characteristics, and frequency-dependent compliance profiles from the comparison of passive and active maps.

10. The method of statement 8, further comprising calibrating for spine-induced pressure artifacts using curvature compensation derived from inertial measurements and a factory-calibrated spine-pressure interaction model.